

Science Olympiad — Cryptalicious Racers - Round 5

Names of participants: (Please print neatly)

School Name:

Team Number:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Question	Value	Incorrect letters	Deduction	Score
1	0			
2	0			
3	0			
4	0			
Final Score				

1) [0 points] During a field trip, you notice these strange markings on a metal plaque. They represent a quote, encoded using an Aristocrat Cipher using a K1 alphabet. What does it say?

JHFW JQO GHKBUGA BTVKJ QJ FTRB QJ JKWIIUGA HG Q

JRQKKVWVY IUEW HZ EWAH IUWRW, XTK U ZWEE HG HGW QGY

UK MQJ MHVJW

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	3			4	2	7	8	4	9	7		2		1		7	3		3	6	3	11	1	2	2
Replacement																										

2) [0 points] A phrase has been encoded using the Porta Cipher using a keyword of BECKON. What does it say?

V	U	W	P	G	X	J	O	O	B	O	O	P	M	G	Q	X	A	R	N	Z	G	H	B	G	W	O	B	L	I

Z	T	B	M	H	U	Y	T	N	A	U	L	R	Q	S	A	M	R	R	U	F	C	G	M	B	N	H	W	X

3) **[0 points]** You came across the following written on a wall in a cave. You notice that it's encoded as a Homophonic using a keyword that has the letters o and s in it. What does it say?

53 29 77 46 17 71 46 2 71 84 94 42 50 71 8 76 53 49 38 31

18 57 69 26 58 14 29 84 86 71 61 40 42 84 65 87 92 49 53 6

61 12 53 31 61 97 65

4) **[0 points]** A riddle is discovered taped under a chemistry lab bench. It's a quote, encrypted as a Baconian. You are told that the cipher text **GOOSE PSYCH IF HAD** decodes to be **ASO**. What does it say?

ABYSS SO SHE MOOSE ANNIE BUGLE AGONY ENTER ELLEN

CLEAR GLASS AT OLD GOOSE PSYCH IF HAD EMILY MOONS

COREY JUICY APPLY CHOPS YOU GO BY THE I STAY JACKS

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Replacement																										